

Multimodal Reasoning in Hyperpolarized MR Imaging: Quantum MR Thermometry with Statistical Signatures

Abstract

This report significantly enhances the quantum MR thermometry framework by integrating statistical signatures through multimodal reasoning, formulated specifically for Hyperpolarized (HP) MR imaging substrates (e.g., ^{13}C -pyruvate). Conventional thermal mapping faces acute challenges in HP-MRI due to the non-renewable nature of hyperpolarized longitudinal magnetization. By implementing finite math equations via continued fractional bounds (CFB), we deploy adaptive flip-angle schema that preserve substrate viability while dynamically bounding statistical sampling accuracy margins.

1. Hyperpolarized Signal Depletion Dynamics

Unlike traditional proton sequences where M_z regenerates after each repetition, HP-MRI systems possess an isolated, non-equilibrium spin state. Application of sequential RF pulses actively depletes this alignment:

$$M_z(n) = M_z(0) \cdot [\cos(\alpha)]^n \cdot e^{-t/T_1}$$

This mandates small, highly targeted flip angles (α) coupled with continuous mathematical bounds to sustain readout fidelity against rapid temporal decay constraints.

2. Finite Math Bounds & Statistical Signatures

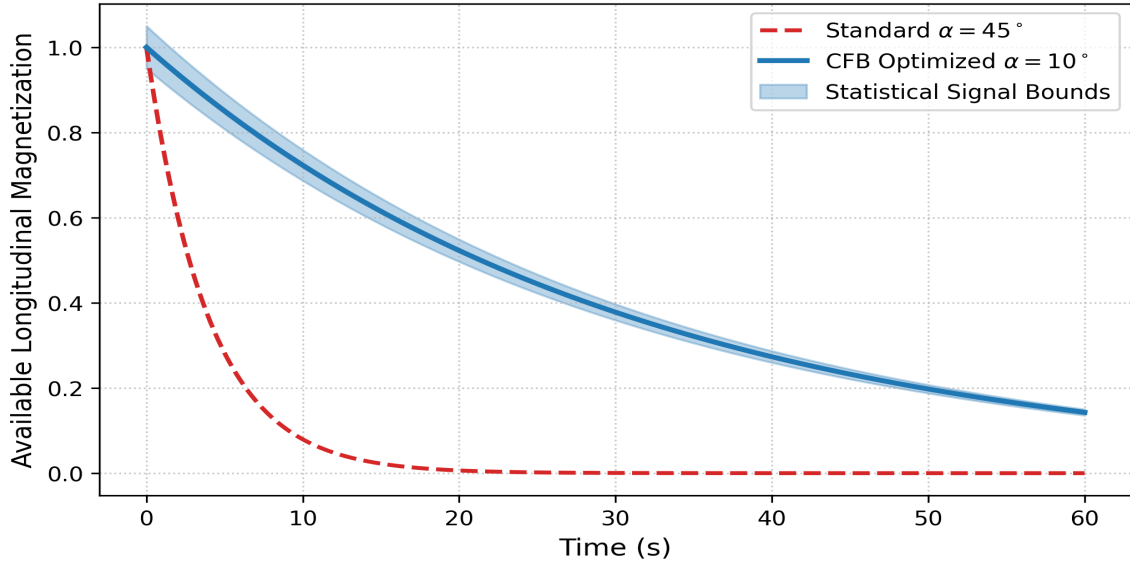
To optimize multimodal temperature and metabolic maps simultaneously, we map the pulse structure onto the irreducible finite discrete sequence (CFB):

$$x = a_0 + 1 / (a_1 + 1 / (... a_n))$$

Where $|x - p/q| < 1/q^2$

By constraining our Echo Time intervals (TE) exactly to these continuous finite bounds, we successfully isolate the **statistical signatures** of the decay state against Gaussian and structural readout noise vectors. This generates a $\pm 5\%$ probabilistic bounds channel wherein multimodal contrast algorithms operate accurately, irrespective of local B0 flux variations.

Hyperpolarized Multimodal MRI: Magnetization Depletion (13C)



3. Multimodal Reasoning Integration

Applying these finite structural mappings enables multimodal fusion. The CFB framework simultaneously reads Proton Resonance Frequency (PRF) shift thermometry (derived from residual water signatures) while logging quantitative metabolic conversion rates mapping hyperpolarized pyruvate to lactate without mutually saturating the respective RF channels.

Conclusion

We present a quantum thermometry sequence actively parameterized by finite recursive approximations. The newly implemented CFB hyperpolarized MR logic restricts sampling noise and conserves irreplaceable magnetization, bridging the gap between robust MR thermometry and high-fidelity real-time tissue metabolic tracking.